

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper: CS-109 — Data Communication

Time: Three hours

Maximum Marks: 70

Instructions: Attempt any five questions. Figures on the right-hand side margin indicate max

Question 1. (a) Define a System. What are the key elements of a system? Distinguish between Conceptual, Physical and Logical System Models. [7]

(a) Explain through a neat diagram the hierarchy of information levels and information systems in an organization. [7]

Question 2. (a) Explain the Software Development Life Cycle (SDLC) model along with its major advantages and disadvantages. [7]

(a) Explain the commonly used Fact Finding Tools for Preliminary Investigation. [7]

Question 3. (a) What is Feasibility Study? Explain the various types of feasibility analysis carried out [7] during system analysis.

(a) What is the purpose of designing SRS document? Explain the typical structuring elements of SRS document. [7]

Question 4. (a) Distinguish between Coupling and Cohesion as design principles. What is desirable [7] and why?

(a) Describe the following terms in reference to System testing: [7] Error, Fault, Failure, Reliability

Question 5. (a) Distinguish between verification & Validation. What are the various levels of [7] software system testing?

(a) What is the major difference between Blackbox and Whitebox testing? Explain the [7] Various criteria for testcase generation in the two approaches.

Question 6. (a) Describe the commonly used methods for System Conversion. [7]

(a) Differentiate between Corrective, Adaptive and Perfective maintenance of a system. [7]

Question 7. Write short notes on any two of the following: [14]

(a) Tangible and Intangible Costs and Benefits

(b) Structured Analysis

(c) Role of a System Analyst

(d) Software Cost Estimation

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